

Math 358: Advanced Calculus Summary
Final Exam Date: 22 April 2026 6.30 - 9.30
Key Results, Theorems, Definitions, etc.
Charles Zitella

Abstract

This document contains a summary of all the key definitions, results, and theorems from class before the final exam. Point-set topology in Euclidean space; continuity and differentiability of functions in several variables. Implicit and inverse function theorems, Lagrange multipliers, Integrals in several variables, Fubini, change of variables, line and surface integrals, Green's theorem, etc...

CONTENTS

1	Continuity and Differentiability	1
1.1	Differentiability	1
1.2	Tangent Planes and Directional Derivatives	3
1.3	Clairaut's Theorem	3
1.4	Inverse Function Theorem and Implicit Function Theorem	4
1.5	Taylor's Theorem	4
1.6	Lagrange Multipliers	5
2	Riemann Integration	6
2.1	Integration over Box domains	6
2.2	Integration over General Domains	7
2.3	Basic Properties of Integrals	8
2.4	Fubini's Theorem	8
2.5	Fubini for General Domains	8
2.6	Change of Variables	10
2.7	Improper Integrals	11
3	Vector Fields, Curves, and Surfaces	12
3.1	Path Integrals/ Line Integrals	12
3.2	Vector Fields and Line Integrals	13
3.3	Conservative Vector Fields	14
3.4	Poincaré's Lemma	14
3.5	Surfaces and Surface Integrals	14
3.6	Green's Theorem	16
3.7	Harmonic Functions	17
3.8	Stokes' Theorem	17
3.9	Divergence Theorem	18
3.10	Divergence Theorem Applications	19

1. CONTINUITY AND DIFFERENTIABILITY

1.1. DIFFERENTIABILITY

Definition 1 (Differentiation). $f : \Omega \rightarrow \mathbb{R}^m$ where $\Omega \subset \mathbb{R}^n$ is a domain¹, $f := (f_1, \dots, f_m)$, $f_i : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is **differentiable** at $x_0 \in \Omega$ if there exists a linear map $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\lim_{x \rightarrow x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0$$

¹Open, Connected

Equivalently, $\forall \epsilon > 0 \exists \delta > 0$ such that for any $x \in \mathbb{R}$ with $\|x - x_0\| < \delta$ ² we have

$$\|f(x) - f(x_0) - L(x - x_0)\| \leq \epsilon \|x - x_0\|$$

Theorem 1. If $f : \Omega \rightarrow \mathbb{R}^m$ is differentiable at $x_0 \in \Omega$, then the linear map L is unique.

Definition 2. If f differentiable at $x_0 \in \mathbb{R}$, with $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ we write the differential map of f at $x_0 \in \Omega$ as $L = Df(x_0)$

Proposition 1. If f is differentiable at $x_0 \in \Omega$, then f is continuous at $x_0 \in \Omega$. In fact, it is Lipschitz continuous at x_0 , i.e. $\exists k > 0$ and $\delta > 0$ such that if $\|x - x_0\| < \delta$, then $\|f(x) - f(x_0)\| \leq k\|x - x_0\|$

Proposition 2. If $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ differentiable at $x_0 \in \Omega$, and $f = (f_1, \dots, f_m)$, $f_i : \Omega \rightarrow \mathbb{R}$, then f differentiable at $x_0 \in \Omega$ iff $f_j : \Omega \rightarrow \mathbb{R}$ is differentiable at $x_0 \in \Omega$ for all $j = 1, \dots, m$.

Definition 3 (Partial Derivatives). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, $f(x_1, \dots, x_m) = (f_1(x_1, \dots, x_n), \dots, f_m(x_1, \dots, x_n))$. Then

$$\frac{\partial f_j}{\partial x_i}(x_0) := \lim_{h \rightarrow 0} \frac{[f_j(x_1, \dots, x_i + h, \dots, x_n) - f_j(x_1, \dots, x_i, \dots, x_n)]}{h}$$

if the limit exists. We call $\frac{\partial f_j}{\partial x_i}(x_0)$ the **partial derivative** of f_j with respect to x_i at x_0 .

Proposition 3. Let $f : \Omega \subset \mathbb{R}^m \rightarrow \mathbb{R}^m$, $f = (f_1, \dots, f_m)$, $f_j : \Omega \rightarrow \mathbb{R}$, be differentiable at $x_0 \in \Omega$. Then all partial derivatives $\frac{\partial f_j}{\partial x_i}(x_0)$ exist and

$$L = Df(x_0) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1}(x_0) & \frac{\partial f_1}{\partial x_2}(x_0) & \dots & \frac{\partial f_1}{\partial x_n}(x_0) \\ \frac{\partial f_2}{\partial x_1}(x_0) & \frac{\partial f_2}{\partial x_2}(x_0) & \dots & \frac{\partial f_2}{\partial x_n}(x_0) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1}(x_0) & \frac{\partial f_m}{\partial x_2}(x_0) & \dots & \frac{\partial f_m}{\partial x_n}(x_0) \end{bmatrix}$$

called the Jacobian of f at x_0 ³.

Remark. f differentiable at $x_0 \in \Omega \Rightarrow \frac{\partial f_j}{\partial x_i}(x_0)$ exists for all i, j . However, the converse need not be true, for example the classic example of $f : \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Both partial derivatives of f at $(0, 0)$ exist and are equal to 0, but f is not differentiable at $(0, 0)$.

Steps to solve: Find candidate for $Df(0, 0)$ using the partial derivatives, then show that the limit in the definition of differentiability does not go to 0 for this candidate.

Theorem 2. Let $f = (f_1, \dots, f_m) : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$. Suppose that all partial derivatives $\frac{\partial f_j}{\partial x_i}$ exist and are continuous at $x_0 \in \Omega$. Then f is differentiable at x_0 .

Definition 4. Suppose $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ has continuous partial derivatives $\frac{\partial f}{\partial x_i}$, $i = 1, \dots, n$ at all points $x \in \Omega$. Then f is continuously differentiable on Ω , and we write $f \in C^1(\Omega)$.

Remark. Continuity of partial derivatives is a sufficient condition for differentiability, but it is not necessary. For example, the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Then $\frac{\partial f}{\partial x}(0, 0) = \frac{\partial f}{\partial y}(0, 0) = 0$, however, along the line $x = y^2$, $x = t, y = t^2$, $\frac{\partial f}{\partial x}(t^2, t) = \frac{1}{2} \neq 0$. But, f is differentiable at $(0, 0)$ with $Df(0, 0) = 0$.

²Here, $\|\cdot\|$ is the standard Euclidean norm on $\mathbb{R}^m$ and $\mathbb{R}^n$.

³also called the derivative of f at $x_0 \in \Omega$

Proposition 4 (Basic Properties of Differentiation).

- (1) $f, g : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ both differentiable at $x_0 \in \Omega$, then so is $F = f + g$. Moreover, $DF(x_0) = Df(x_0) + Dg(x_0)$.
- (2) $f, g : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ both differentiable at $x_0 \in \Omega$, then so is $F = f \cdot g : \Omega \rightarrow \mathbb{R}$. Moreover, $DF(x_0) = f(x_0) \cdot Dg(x_0) + g(x_0) \cdot Df(x_0)$.
- (3) $f, g : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ both differentiable at $x_0 \in \Omega$, and $g(x_0) \neq 0$, then so is $F = \frac{f}{g} : \Omega \rightarrow \mathbb{R}$. Moreover,

$$DF(x_0) = \frac{g(x_0) \cdot Df(x_0) - f(x_0) \cdot Dg(x_0)}{g(x_0)^2}$$

- (4) $f : \Omega \subset \mathbb{R}^n \rightarrow \tilde{\Omega} \subset \mathbb{R}^m$, $g : \tilde{\Omega} \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$ with f differentiable at $x_0 \in \Omega$, and g differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$. Then $H = g \circ f$ is differentiable at $x_0 \in \Omega$ and

$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

Theorem 3 (Chain Rule). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \tilde{\Omega} \subset \mathbb{R}^m$, $\tilde{\Omega} = f(\Omega)$, $g : \tilde{\Omega} \subset \mathbb{R}^m \rightarrow \mathbb{R}^k$. If f is differentiable at $x_0 \in \Omega$ and g is differentiable at $y_0 = f(x_0) \in \tilde{\Omega}$, then $H = g \circ f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^k$ is differentiable at $x_0 \in \Omega$, and

$$DH(x_0) = Dg(y_0) \cdot Df(x_0)$$

Example 1.1. Given f differentiable in $\mathbb{R}^2$ and $g(r, \theta) = (r \cos \theta, r \sin \theta)$, with $(r, \theta) \in (0, \infty) \times [0, 2\pi]$. Set $F(r, \theta) = f(g(r, \theta))$, compute $\frac{\partial F}{\partial r}(r, \theta)$ and $\frac{\partial F}{\partial \theta}(r, \theta)$ in terms of the partial derivatives of f , using the chain rule.

To solve, the multivariable chain rule gives us

$$\frac{\partial F}{\partial r} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial r} = \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta$$

and

$$\frac{\partial F}{\partial \theta} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial \theta} = -r \frac{\partial f}{\partial x} \sin \theta + r \frac{\partial f}{\partial y} \cos \theta$$

1.2. TANGENT PLANES AND DIRECTIONAL DERIVATIVES

Definition 5 (Gradient). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on Ω . Then $Df(x) = (\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x)) = \nabla f(x)$ is called the **gradient** of f at $x \in \Omega$.

Definition 6 (Tangent Planes). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on Ω . $S := \text{graph } f = \{(x, z) \in \Omega \times \mathbb{R} : z = f(x)\}$. Then for $x_0 \in \Omega$, $T_{x_0}S = \{(x, z) \in \mathbb{R}^n \times \mathbb{R} : z = f(x_0) + Df(x_0) \cdot (x - x_0)\}$

Definition 7 (Directional Derivative). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable on Ω . Then for $x_0 \in \Omega$ and $v \in \mathbb{R}^n$, the directional derivative of f at x_0 in the direction of v is defined as

$$D_v f(x_0) = \langle \nabla f(x_0), v \rangle$$

Proposition 5. Suppose $\nabla f(x) \neq 0$. Then $\nabla f(x)$ points in the direction of the steepest increase of f .⁴

⁴Proof follows from the Cauchy-Schwarz inequality, $\langle a, b \rangle = \|a\| \cdot \|b\| \cos \theta$

1.3. CLAIRAUT'S THEOREM

Theorem 4 (MVT in $\mathbb{R}^n$). Let $B \subset \mathbb{R}^n$ be a ball, $B = \{x \in \mathbb{R}^n : \|x - x_0\| < p \text{ for some } x_0\}$, and $f : B \rightarrow \mathbb{R}$ differentiable on B . Then, for any $x, y \in B$ there exists $z \in B$ such that

$$f(y) - f(x) = Df(z) \cdot (y - x)$$

As a consequence, $\|f(y) - f(x)\| \leq \|Df(z)\| \cdot \|y - x\|$.

Definition 8. We say f twice differentiable at $x \in \Omega$ if Df exists locally near $x \in \Omega$ and $Df : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is differentiable at $x \in \Omega$. Then $D^2f = D(Df)$ at $x \in \Omega$. Similarly, for all $k \geq 2$, $D^k f = D(D^{k-1}f)$.

Definition 9. Given $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, we say that $f \in C^k(\Omega)$ if all the partial derivatives up to order k are continuous on Ω .

Definition 10. (Hessian) Given $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, the **Hessian** matrix of f at $x \in \Omega$ is

$$Hf(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(x) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(x) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(x) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(x) & \frac{\partial^2 f}{\partial x_n \partial x_2}(x) & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n}(x) \end{bmatrix}$$

For $f(x, y)$, $Df = (\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ and

$$D^2f = Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x \partial x} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y \partial y} \end{bmatrix}$$

Theorem 5 (Clairaut's Theorem). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable at $x \in \Omega$. Then,

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$$

for all $i, j = 1, \dots, n$. Moreover, if $f \in C^2(\Omega)$, then the above holds for all $x \in \Omega$.

Corollary 1. If $\frac{\partial^2 f}{\partial x_i \partial x_j}(x)$ are all continuous at $x \in \Omega$ for all $i, j = 1, \dots, n$, then $\frac{\partial^2 f}{\partial x_i \partial x_j}(x) = \frac{\partial^2 f}{\partial x_j \partial x_i}(x)$ for all $i, j = 1, \dots, n$.

1.4. INVERSE FUNCTION THEOREM AND IMPLICIT FUNCTION THEOREM

Theorem 6 (Banach Fixed Point). Let (X, d) be a complete metric space, and $f : X \rightarrow X$ be a contraction mapping with $d(f(x), f(y)) \leq \alpha d(x, y)$ for some $0 < \alpha < 1$. Then $\exists$ unique $x_0 \in X$ such that $f(x_0) = x_0$.

Definition 11. Let $M_n := \{n \times n \text{ matrices}\} \cong \mathbb{R}^{n^2}$ and $\|A\| := \sqrt{\sum_{i,j} a_{ij}^2}$ for $A = (a_{ij}) \in M_n$.

Let $GL_n := \{A \in M_n : \det A \neq 0\} = \det^{-1}(\mathbb{R} \setminus \{0\})$. Thus since $\det : M_n \rightarrow \mathbb{R}$ is continuous, GL_n is an open subset of M_n .

Theorem 7 (Inverse Function). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 . Let $x_0 \in \Omega$ and assume $Df(x_0) \in GL_n$. Then there exists neighbourhoods U and V of x_0 and $f(x_0)$ respectively such that $f(U) = V$ and $f|_U$ has a C^1 inverse map $f^{-1} : V \rightarrow U$. Moreover, for any $y \in V$, and $x = f^{-1}(y)$, we have $Df^{-1}(y) = [Df(x)]^{-1}$.

Remark. If $f \in C^k$, $k \geq 1$, then the inverse map f^{-1} is also in C^k .

Example 1.2. IFT is local. For example, $f(x, y) = (e^y \cos x, e^y \sin x)$. Then $Df(x, y) = \begin{bmatrix} -e^y \sin x & e^y \cos x \\ e^y \cos x & e^y \sin x \end{bmatrix}$, and $\det Df(x, y) = -(\cos^2 x + \sin^2 x)e^{2y} = -e^{2y} \neq 0$ for all $(x, y) \in \mathbb{R}^2$. However, $f(x + 2k\pi, y) = f(x, y)$ for all $k \in \mathbb{Z}$, so f is not injective globally on (x, y) .

Theorem 8 (Implicit Function Theorem). $F : \Omega \subset \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ C^k map, $X = (x, y) \in \mathbb{R}^n \times \mathbb{R}^m$ and $X_0 = (x_0, y_0)$ with $F(X_0) = 0$. Writing $F = (F_1, \dots, F_m)$, assume that

$$D_y F(X_0) = \begin{bmatrix} \frac{\partial F_1}{\partial y_1} & \frac{\partial F_1}{\partial y_2} & \cdots & \frac{\partial F_1}{\partial y_m} \\ \frac{\partial F_2}{\partial y_1} & \frac{\partial F_2}{\partial y_2} & \cdots & \frac{\partial F_2}{\partial y_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial y_1} & \frac{\partial F_m}{\partial y_2} & \cdots & \frac{\partial F_m}{\partial y_m} \end{bmatrix} (X_0) \in GL_m$$

Then there exists neighbourhoods U and V of $x_0 \in \mathbb{R}^n$ and $y_0 \in \mathbb{R}^m$ and a unique C^k map $f : U \rightarrow V$ such that $F(x, f(x)) = 0$, for all $x \in U$.

Corollary 2. Let $F : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be $C^k(\Omega)$. Let $x = (x', y) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and suppose $(x'_0, y_0) \in \Omega$ with $\frac{\partial f}{\partial y}(x'_0, y_0) \neq 0$. Then there exists neighbourhoods U, V of $x'_0 \in \mathbb{R}^{n-1}$ and $y_0 \in \mathbb{R}$ and a unique function $f : U \rightarrow V$, $f \in C^k(U)$ such that $\{F(x', y) = 0\} = \{y = f(x')\} \forall x' \in U$.

Theorem 9 (Morse Lemma). $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ a C^k function $k \geq 3$. Let $0 \in \Omega$ be a critical point, i.e. $\nabla f(0) = 0$, and assume $f(0) = 0, \nabla f(0) = 0, Hf(0) \in GL_n$. Then there exists open sets U and V of $0 \in U \cap V$ and $g \in C^{k-2}(U)$, $g : U \rightarrow V$ with $g^{-1} : U \rightarrow V$, $g^{-1} \in C^2(V)$ such that $f(g(y)) = y_{\ell+1}^2 + \dots + y_n^2 - (y_1^2 + y_2^2 + \dots + y_\ell^2)$. Here $\ell \in \mathbb{Z}$, $\ell \in [0, n]$.

1.5. TAYLOR'S THEOREM

Theorem 10 (Taylor's Theorem in $\mathbb{R}^n$). Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^{k+1}(\Omega)$. Then, for any $x^0 \in \Omega$,

$$f(x) = f(x^0) + \sum_{j=1}^k \frac{1}{j!} \left[\sum_{i_1=1}^n \cdots \sum_{i_j=1}^n \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0) (x_{i_1} - x_{i_1}^0) \dots (x_{i_j} - x_{i_j}^0) \right] + R_k(f)(x)$$

where $\frac{|R_k(f)(x)|}{\|x - x^0\|^k} \leq M_k \|x - x^0\|$ as $x \rightarrow x^0$.

Remark. The k -th order Taylor polynomial of f at x^0 is the sum of the first two terms in the above expansion, i.e.

$$P_k(f)(x) = f(x^0) + \sum_{j=1}^k \frac{1}{j!} \left[\sum_{i_1=1}^n \cdots \sum_{i_j=1}^n \frac{\partial^j f}{\partial x_{i_1} \dots \partial x_{i_j}}(x^0) (x_{i_1} - x_{i_1}^0) \dots (x_{i_j} - x_{i_j}^0) \right]$$

This is a good approximation to $f(x)$ near x^0 provided $\lim_{x \rightarrow x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0$.

Theorem 11. Let $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be $C^{k-1}(\Omega)$ and assume $D^{k-1}f$ is differentiable at $x^0 \in \Omega$. Then the k -th order Taylor expansion in the previous theorem still holds, but with the only assumption of the remainder term being $\lim_{x \rightarrow x^0} \frac{|R_k(f)(x)|}{\|x - x^0\|^k} = 0$.

Example 1.3. Let $f(x, y) = e^x + \sin(xy)$. Compute the second order Taylor expansion of f around $(0, 0)$.

To solve this problem, we first recognize that $f \in C^\infty(\mathbb{R}^2)$, so we can apply Taylor's theorem. We compute the first and second order partial derivatives of f at $(0, 0)$:

$$f_x = \frac{\partial f}{\partial x} = e^x + y \cos(xy), \quad f_y = \frac{\partial f}{\partial y} = x \cos(xy)$$

Therefore $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$. Next, we compute the second order partial derivatives:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2} = e^x - y^2 \sin(xy), \quad f_{yy} = \frac{\partial^2 f}{\partial y^2} = -x^2 \sin(xy), \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = \cos(xy) - xy \sin(xy)$$

Therefore, $f_{xx}(0, 0) = 1$, $f_{yy}(0, 0) = 0$, and $f_{xy}(0, 0) = 1$. Thus, the second order Taylor expansion of f around $(0, 0)$ is

$$f(x, y) = f(0, 0) + f_x(0, 0)x + f_y(0, 0)y + \frac{1}{2!} [f_{xx}(0, 0)x^2 + 2f_{xy}(0, 0)xy + f_{yy}(0, 0)y^2] + R_2(f)(x, y)$$

which simplifies to

$$f(x, y) = 1 + x + \frac{x^2}{2} + xy + R_2(f)(x, y)$$

and the remainder term satisfies $\frac{|R_2(f)(x, y)|}{x^2 + y^2} \leq M\sqrt{x^2 + y^2}$ as $(x, y) \rightarrow (0, 0)$.

1.6. LAGRANGE MULTIPLIERS

Theorem 12 (Lagrange Multiplier). Let $f, g : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be C^1 functions in $\mathbb{R}$. Let $\Sigma = \{x \in \Omega : g(x) = 0\}$ and assume that $Dg(x_0) \neq 0$ for all $x_0 \in \Sigma$. Then if $f|_\Sigma$ has a max or min at $x_0 \in \Sigma$, then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0)$$

Theorem 13 (Max/Min). Let $\tilde{\Omega} \supset \Omega$ be a slightly larger domain and assume $f, g \in C^1(\Omega)$, where $\partial\Omega = \{x \in \tilde{\Omega} : g(x) = 0\}$, $\nabla g(x) \neq 0 \forall x \in \partial\Omega$. Then,

- (1) If $x_0 \in \partial\Omega$ is a local max/min of $f|_{\partial\Omega}$, then $\nabla f(x_0) = \lambda \nabla g(x_0)$ for some $\lambda \in \mathbb{R}$.
- (2) If $x_0 \in \text{Int}(\Omega)$ is a local max/min of f in $\text{Int}(\Omega)$, then $\nabla f(x_0) = 0$.
- (3) Global max/min of $f|_{\tilde{\Omega}}$ is largest/smallest value of f at all points in 1 and 2.

Theorem 14 (Multiple Constraints). $f : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $g_i : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, k$ are C^1 functions. Let $\Gamma_0 = \{x \in \Omega : g_i(x) = 0, i = 1, \dots, k\}$ and assume that $\{\nabla g_1(x), \dots, \nabla g_k(x)\}$ are linearly independent for all $x \in \Gamma_0$. Then if $f|_{\Gamma_0}$ has a max/min at $x_0 \in \Gamma_0$, then there exists $\lambda_1, \dots, \lambda_k \in \mathbb{R}$ such that

$$\nabla f(x_0) = \sum_{i=1}^k \lambda_i \nabla g_i(x_0)$$

Example 1.4. Find the absolute max and min of $f(x_1, \dots, x_n) = x_1 + x_2$ on $B_1(0) = \{|x|^2 \leq 1\} = \{x_1^2 + \dots + x_n^2 \leq 1\}$, $\partial B_1(0) = \{x_1^2 + \dots + x_n^2 = 1\}$.

To solve, first observe $f \in C^k(\mathbb{R}^n)$ for all $k \geq 1$. Thus we will apply the max/min theorem with $g(x) = |x|^2 - 1 = x_1^2 + \dots + x_n^2 - 1$. Then, since $\nabla f(x) = (1, 1, 0, \dots, 0) \neq 0$, there are no critical points in $\text{Int}(B_1(0)) = \{|x|^2 < 1\}$. So the global max/min must occur on $\partial B_1(0)$.

By Lagrange multiplier theorem, at max/min $x_0 \in \partial B_1(0)$, there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0) \Leftrightarrow (1, 1, 0, \dots, 0) = \lambda(2x_1^0, 2x_2^0, 2x_3^0, \dots, 2x_n^0) \quad (1)$$

$$\Leftrightarrow x_1^0 = x_2^0 = \frac{1}{2\lambda}, x_3^0 = \dots = x_n^0 = 0 \quad (2)$$

Also,

$$(x_1^0)^2 + (x_2^0)^2 = 1 \Leftrightarrow \frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1 \Leftrightarrow \lambda = \pm \frac{1}{\sqrt{2}}$$

Thus, the candidates for max/min are $x_0 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right)$ and $x_0 = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right)$. Evaluating f at these points, we get

$$f\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0\right) = \sqrt{2}, \quad f\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, \dots, 0\right) = -\sqrt{2}$$

Thus, the absolute max of f on $B_1(0)$ is $\sqrt{2}$ and the absolute min is $-\sqrt{2}$.

2. RIEMANN INTEGRATION

2.1. INTEGRATION OVER BOX DOMAINS

Definition 12 (Partition). Let $B = [a_1, b_1] \times \dots \times [a_n, b_n]$. A **partition** of B is an ordered set

$$\mathcal{P} = \{P_1 = \{x_1^0, x_1^1, \dots, x_1^{m_1}\}, \dots, P_n = \{x_n^0, x_n^1, \dots, x_n^{m_n}\}\} \quad ^5$$

Here, for any given j , P_j is the set of x_j^i partition of $[a_j, b_j]$ s.t.

$$a_j = x_j^0 < x_j^1 < \dots < x_j^{m_j-1} < x_j^{m_j} = b_j$$

Definition 13 (Refinement). Given two partitions $\mathcal{P}$ and $\tilde{\mathcal{P}}$, we say that $\tilde{\mathcal{P}}$ is a **refinement** of $\mathcal{P}$, i.e. $\tilde{\mathcal{P}} \supset \mathcal{P}$, if for each $j = 1, \dots, n$,

$$P_j = \{x_j^0, x_j^1, \dots, x_j^{m_j}\} \subset \{\tilde{x}_j^0, \tilde{x}_j^1, \dots, \tilde{x}_j^{m_j}\} = \tilde{P}_j$$

Definition 14 (Sub-box). Given partition $\mathcal{P}$ of B , $\mathcal{P} = \{P_1, \dots, P_j\}$, let

$$B_{i_1, i_2, \dots, i_n} = [x_1^{i_1-1}, x_1^{i_1}] \times \dots \times [x_n^{i_n-1}, x_n^{i_n}]$$

be a **sub-box** of B for some $\{i_1, \dots, i_n\}$ partition indices, i.e. $1 \leq i_j \leq m_j$ for all $j = 1, \dots, n$. Then define the length of the j th side of the sub-box as

$$\Delta x_j^{i_j} := x_j^{i_j} - x_j^{i_j-1} \quad \forall j$$

Thus the volume is intuitively defined as

$$\Delta V_{i_1, \dots, i_n} := \Delta x_1^{i_1} \cdot \dots \cdot \Delta x_n^{i_n} = \prod_{j=1}^n (x_j^{i_j} - x_j^{i_j-1})$$

Definition 15 (Upper and Lower Riemann Sums). Let $f : B \rightarrow \mathbb{R}$ be bounded and $\mathcal{P}$ a partition of B . Define the **lower** and **upper** Riemann sums of f with respect to $\mathcal{P}$ as

$$L(f, \mathcal{P}) = \sum_{i_1=1}^{m_1} \dots \sum_{i_n=1}^{m_n} \left(\inf_{x \in B_{i_1, \dots, i_n}} f(x) \right) \cdot \Delta V_{i_1, \dots, i_n} \quad \text{and} \quad U(f, \mathcal{P}) = \sum_{i_1=1}^{m_1} \dots \sum_{i_n=1}^{m_n} \left(\sup_{x \in B_{i_1, \dots, i_n}} f(x) \right) \cdot \Delta V_{i_1, \dots, i_n}$$

Lemma 15. Let $f : B \rightarrow \mathbb{R}$ be bounded, and $\tilde{\mathcal{P}} \supset \mathcal{P}$ be a refinement. Then

$$L(f, \mathcal{P}) \leq L(f, \tilde{\mathcal{P}}) \leq U(f, \tilde{\mathcal{P}}) \leq U(f, \mathcal{P})$$

Definition 16. Let $f : B \rightarrow \mathbb{R}$ be bounded, i.e. $\exists m, M$ s.t. $m \leq f(x) \leq M$, $\forall x \in \mathbb{R}$. Then define

$$U(f) := \inf_{\mathcal{P}} U(f, \mathcal{P}) \quad \text{and} \quad L(f) := \sup_{\mathcal{P}} L(f, \mathcal{P})$$

⁵The lower index denotes the dimension, and the upper index denotes the partition index.

Definition 17. Let $f : B \rightarrow \mathbb{R}$ be bounded. Then we say f **Riemann integrable** over B provided $L(f) = U(f)$, and we define

$$\int_B f(x) dV := L(f) = U(f)$$

Lemma 16. Let $f : B \rightarrow \mathbb{R}$ be bounded. Then f is Riemann integrable over B iff $\forall \epsilon > 0, \exists \mathcal{P}$ partition of B such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) \leq \epsilon$.

Theorem 17. Let $f : B \rightarrow \mathbb{R}$ be $C(B)$. Then f is Riemann integrable over B . (Continuity implies integrability)

Lemma 18 (MVT for Integrals). Let $f : B \rightarrow \mathbb{R}$ be continuous. Then $\exists x_0 \in B$ s.t. $\int_B f(x) dV = f(x_0) \cdot \text{Vol}(B)$, where $\text{Vol}(B) = \int_B 1 dV$.

2.2. INTEGRATION OVER GENERAL DOMAINS

Definition 18. Given $\Omega \subset \mathbb{R}^n$ bounded, $f : \Omega \rightarrow \mathbb{R}$ bounded, we let $B \subset \mathbb{R}^n$ be a box. We define $F : B \rightarrow \mathbb{R}$ as

$$F(x) := \begin{cases} f(x) & x \in \Omega \\ 0 & x \in B \setminus \Omega \end{cases}$$

Then

$$\int_{\Omega} f dV = \int_B F dV \quad \text{when the latter exists}$$

Definition 19 (Content Zero). $\Gamma \subset \mathbb{R}^n$ is of **Content 0** if $\forall \epsilon > 0$, there exists open box domains $B_1^0, \dots, B_N^0$ s.t.

- (1) $\Gamma \subset \bigcup_{i=1}^N B_i^0$
- (2) $\sum_{i=1}^N \text{Vol}(B_i^0) < \epsilon$

Example 2.1. If $\Gamma_1, \dots, \Gamma_n$ are all content 0, then $\text{Content}(\Gamma_1 \cup \dots \cup \Gamma_N) = 0$

To solve, let $\epsilon > 0$. Since Γ_i is of content 0 for all $i = 1, \dots, N$, there exists open box domains $B_1^i, \dots, B_{M_i}^i$ s.t. $\Gamma_i \subset \bigcup_{j=1}^{M_i} B_j^i$ and $\sum_{j=1}^{M_i} \text{Vol}(B_j^i) < \frac{\epsilon}{N}$ for all i . Then we have

$$\Gamma_1 \cup \dots \cup \Gamma_N \subset \bigcup_{i=1}^N \bigcup_{j=1}^{M_i} B_j^i$$

and

$$\sum_{i=1}^N \sum_{j=1}^{M_i} \text{Vol}(B_j^i) < \sum_{i=1}^N \frac{\epsilon}{N} = \epsilon$$

Theorem 19. Suppose $f : B \subset \mathbb{R}^n \rightarrow \mathbb{R}$ bounded. Let $\Gamma(f)$ = discontinuity set of f .

Then $\text{Content}(\Gamma(f)) = 0$ iff f is Riemann integrable over B .

Example 2.2. $\Omega \subset \mathbb{R}^{n+1}$ compact. $g : \Omega \rightarrow \mathbb{R}$ continuous. Then the graph of g , $\Gamma = \{(x', g(x')) : x' \in \Omega\}$, satisfies $\text{Content}(\Gamma) = 0$.

To solve,

Corollary 3. Suppose $f : B \rightarrow \mathbb{R}$ is bounded and $\Gamma(f)$ is the union of finitely many graphs of continuous functions over closed domains in $\mathbb{R}^{n-1}$. Then f is Riemann integrable over B .

Corollary 4. $\Omega \subset \mathbb{R}^n$ closed and bounded, $\partial\Omega$ is the union of finitely many graphs of C^0 functions over closed domains in $\mathbb{R}^{n-1}$. Suppose $f : \Omega \rightarrow \mathbb{R}$ bounded and $\text{Content}(\Gamma(f) \cap \Omega) = 0$. Then f is Riemann integrable over Ω .

Corollary 5. Suppose Ω closed and bounded, $\Omega \subset \tilde{\Omega}$ with $G \in C^1(\tilde{\Omega})$ s.t. $\Omega = \{G(x) \leq 0 : x \in \tilde{\Omega}\}$ and $\partial\Omega = \{G(x) = 0\}$ with $\nabla G(x) \neq 0$ for all $x \in \partial\Omega$. Suppose $f : \Omega \rightarrow \mathbb{R}$ is bounded and $\text{Content}(\Gamma(f) \cap \Omega) = 0$. Then f is Riemann integrable over Ω .

2.3. BASIC PROPERTIES OF INTEGRALS

Proposition 6. Assume $\Omega \subset \mathbb{R}^n$ bounded and $f, g : \Omega \rightarrow \mathbb{R}$ are both Riemann integrable over Ω . Then,

- (1) $\int_{\Omega} (f + g) dV = \int_{\Omega} f dV + \int_{\Omega} g dV$;
- (2) $\int_{\Omega} \alpha f dV = \alpha \int_{\Omega} f dV$ for any $\alpha \in \mathbb{R}$;
- (3) $f \leq g \Rightarrow \int_{\Omega} f dV \leq \int_{\Omega} g dV$;
- (4) If $\Omega = \Omega_1 \cup \Omega_2$, Ω bounded and $\text{Content}(\Omega_1 \cap \Omega_2) = 0$. Then $\int_{\Omega} f dV = \int_{\Omega_1} f dV + \int_{\Omega_2} f dV$.

2.4. FUBINI'S THEOREM

Definition 20. Given $f : B \rightarrow \mathbb{R}$ bounded, we define the iterated integrals as

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \left(\int_c^d f(x, y) dy \right) dx \quad \text{and} \quad \int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy$$

Theorem 20 (Fubini for box domains). Suppose $f : B \rightarrow \mathbb{R}$ is Riemann integrable over B , where $B = [a, b] \times [c, d]$.

- (1) Assume $\int_c^d f(x, y) dy$ exists $\forall x \in [a, b]$. Then $\int_a^b \int_c^d f(x, y) dy dx$ exists and equals $\int_B f(x, y) dV$.
- (2) Assume $\int_a^b f(x, y) dx$ exists $\forall y \in [c, d]$. Then $\int_c^d \int_a^b f(x, y) dx dy$ exists and equals $\int_B f(x, y) dV$.

2.5. FUBINI FOR GENERAL DOMAINS

Definition 21 (y -simple, x -simple domains). A domain $\Omega \subset \mathbb{R}^2$ is called **y -simple** if it is of the form

$$\Omega = \{a \leq x \leq b : \varphi_1(x) \leq y \leq \varphi_2(x)\}$$

where $\varphi_1, \varphi_2 \in C^0([a, b])$.

Similarly, $\Omega \subset \mathbb{R}^2$ is called **x -simple** if it is of the form

$$\Omega = \{c \leq y \leq d : \psi_1(y) \leq x \leq \psi_2(y)\}$$

where $\psi_1, \psi_2 \in C^0([c, d])$. $\Omega \subset \mathbb{R}^2$ is called **simple** if it is both y -simple and x -simple.

Theorem 21 (Fubini). Suppose $\Omega \subset \mathbb{R}^2$ bounded and f integrable over Ω . Then,

- (1) If Ω is y -simple and $\forall x \in [a, b]$, $\int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy$ exists, then

$$\int_{\Omega} f(x, y) dV = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx$$

- (2) If Ω is x -simple and $\forall y \in [c, d]$, $\int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx$ exists, then

$$\int_{\Omega} f(x, y) dV = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy$$

If Ω is simple, then both hold.

Definition 22. $\Omega \subset \mathbb{R}^2$ is **elementary** if Ω is x -simple or y -simple.

Example 2.3. Compute $\iint_{\Omega} x \, dA$ where $\Omega = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq \sqrt{\pi}, 0 \leq y \leq \sin x^2\}$

To solve, first note that $f(x, y) = x$ is C^0 on Ω . Since Ω is bounded, $\iint_{\Omega} x \, dV$ exists. Also, Ω is y -simple with $\varphi_1(x) = 0, \varphi_2(x) = \sin x^2, \varphi_1, \varphi_2 \in C^0([0, \sqrt{\pi}])$. By Fubini,

$$\iint_{\Omega} x \, dA = \int_0^{\sqrt{\pi}} \int_0^{\sin x^2} x \, dy \, dx = \int_0^{\sqrt{\pi}} x \sin x^2 \, dx = \frac{1}{2} \int_0^{\pi} \sin u \, du = \frac{1}{2} [-\cos u]_0^{\pi} = 1$$

Example 2.4. Compute $I = \int_0^1 \int_y^1 e^{x^2} \, dx \, dy$.

To solve, first note $\int_y^1 e^{x^2} \, dx$ doesn't exist in closed-form. But since $e^{x^2} \in C^0(\Omega)$, $\iint_{\Omega} e^{x^2} \, dA$ exists. Since Ω is simple, by Fubini,

$$\iint_{\Omega} e^{x^2} \, dA = \int_0^1 \int_0^x e^{x^2} \, dy \, dx = \int_0^1 x e^{x^2} \, dx = \frac{1}{2} \int_0^1 e^u \, du = \frac{1}{2}(e - 1)$$

Definition 23. Some notation: Given a box domain $B \subset \mathbb{R}^n$, write

$$\int_B f \, dV := L(f) = \sup_{\mathcal{P}} L(f, \mathcal{P}) \quad \text{and} \quad \overline{\int_B f \, dV} := U(f) = \inf_{\mathcal{P}} U(f, \mathcal{P})$$

Theorem 22. Let $B = B_1 \times B_2$ box domain where $B_1 \subset \mathbb{R}^k$ and $B_2 \subset \mathbb{R}^{n-k}$ also a box. Suppose $f : B \rightarrow \mathbb{R}$ is Riemann integrable. Set

$$\ell(x) = \int_{\underline{B_2}} f(x, y) \, dV(y) \quad u(x) = \overline{\int_{B_2} f(x, y) \, dV(y)} \quad \forall x \in B_1 \subset \mathbb{R}^k$$

and

$$\tilde{\ell}(y) = \int_{\underline{B_1}} f(x, y) \, dV(x) \quad \tilde{u}(y) = \overline{\int_{B_1} f(x, y) \, dV(x)} \quad \forall y \in B_2 \subset \mathbb{R}^{n-k}$$

Then $\ell(x), u(x)$ are Riemann integrable over B_1 . Also $\tilde{\ell}(y), \tilde{u}(y)$ are Riemann integrable over B_2 . Moreover,

$$\begin{aligned} \int_B f(x, y) \, dV &= \int_{B_1} \ell(x) \, dV(x) = \int_{B_1} u(x) \, dV(x) \\ \int_B f(x, y) \, dV &= \int_{B_2} \tilde{\ell}(y) \, dV(y) = \int_{B_2} \tilde{u}(y) \, dV(y) \end{aligned}$$

Definition 24 (z -simple). A bounded domain $W \subset \mathbb{R}^3$ is called **z -simple** if $\exists$ an elementary region $\Omega \subset \mathbb{R}^2$ and $\eta_1, \eta_2 \in C^0(\Omega)$ s.t.

$$W = \{(x, y, z) \in \mathbb{R}^3 : (x, y) \in \Omega \text{ and } \eta_1(x, y) \leq z \leq \eta_2(x, y)\}$$

It is possible to permute the coordinates and define x -simple and y -simple domains in $\mathbb{R}^3$ similarly.

Theorem 23 (Fubini in $\mathbb{R}^3$). Suppose $W \subset \mathbb{R}^3$ is bounded, z -simple, and $f : W \rightarrow \mathbb{R}$ is Riemann integrable over W . Assume $\forall (x, y) \in \Omega$,

$$\int_{\eta_1(x, y)}^{\eta_2(x, y)} f(x, y, z) \, dz \quad \text{exists}$$

Then it is Riemann integrable over Ω , and

$$\iiint_W f \, dV = \int_{\Omega} \left(\int_{\eta_1(x, y)}^{\eta_2(x, y)} f(x, y, z) \, dz \right) dV(x, y)$$

The same holds for x -simple and y -simple.

2.6. CHANGE OF VARIABLES

Definition 25 (Regular Partition). Let $\Omega \subset \mathbb{R}^n$ be a domain with $\text{Content}(\partial\Omega) = 0$. A **regular partition**, $\mathcal{P}_{reg} = \{D_k\}_{k=1}^N$ is a collection of subdomains $D_k \subset \Omega$; $k = 1, \dots, N$ s.t.

$$(1) \Omega = \bigcup_{k=1}^N D_k, \text{Content}(\partial D_k) = 0 \quad \forall k = 1, \dots, N.$$

$$(2) \text{Content}(D_k \cap D_\ell) = 0 \quad \forall k \neq \ell.$$

Definition 26. Suppose $f : \Omega \rightarrow \mathbb{R}$ is bounded, where $\Omega \subset \mathbb{R}^n$ is a bounded domain, and $\mathcal{P}_{reg} = \{D_k\}_{k=1}^N$. Define

$$U(f, \mathcal{P}_{reg}) = \sum_{k=1}^N (\sup_{D_k} f) \cdot \text{Vol}(D_k) \quad L(f, \mathcal{P}_{reg}) = \sum_{k=1}^N (\inf_{D_k} f) \cdot \text{Vol}(D_k)$$

Lemma 24. $f : B \subset \mathbb{R}^n \rightarrow \mathbb{R}$ is Riemann integrable over box B iff $\forall \epsilon > 0$, $\exists \mathcal{P}_{reg}$ regular partition of B s.t.

$$U(f, \mathcal{P}_{reg}) - L(f, \mathcal{P}_{reg}) \leq \epsilon$$

Theorem 25 (Change of Variables). Let $\Omega, \Omega' \subset \mathbb{R}^n$ be bounded domains. Let $T : \Omega' \rightarrow \Omega$ be a C^1 map that is bijective from Ω' onto Ω . Then, for any $f : \Omega \rightarrow \mathbb{R}$ Riemann integrable over Ω ,

$$\int_{\Omega} f \, dV = \int_{\Omega'} (f \circ T) \cdot |\det DT| \, dV$$

Example 2.5 (Polar Coordinates). Let $x = r \cos \theta$ and $y = r \sin \theta$. $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq a^2\}$, $\Omega' = \{(r, \theta) \in (0, a] \times [0, 2\pi)\}$. Then $T : \Omega' \rightarrow \Omega$ defined by $T(r, \theta) = (r \cos \theta, r \sin \theta)$ is a C^1 bijection.

Also, $DT = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$ and $\det DT = r$. Thus, for any $f : \Omega \rightarrow \mathbb{R}$ Riemann integrable over Ω ,

$$\iint_{\{x^2+y^2 \leq a^2\}} f(x, y) \, dA = \int_0^a \int_0^{2\pi} f(r \cos \theta, r \sin \theta) \cdot r \, d\theta \, dr$$

Remark. Change of variables still holds if we only assume $T : \Omega' \setminus \Gamma' \rightarrow \Omega$ is a C^1 bijection, where $\text{Content}(\Gamma') = 0$

Example 2.6. Compute $\iint_{\Omega} (y - x) \cos xy \, dA$ where Ω is the region defined by $x + y = 1, x + y = 4, xy = -1, xy = -4, y \geq x$.

To solve, first note that $f(x, y) = (y - x) \cos xy$ is C^0 on Ω , thus it is Riemann integrable over Ω . Now to solve, we will perform a change of variables with $u = x + y$ and $v = xy$. Then $T : \Omega' \rightarrow \Omega$ defined by $T(u, v) = (x, y)$ is a C^1 bijection, where $\Omega' = \{(u, v) : 1 \leq u \leq 4, -4 \leq v \leq -1\}$. Also, $|\det(DT^{-1})| |\det \begin{bmatrix} 1 & 1 \\ y & x \end{bmatrix}| = |x - y| = \sqrt{u^2 - 4v} = \frac{1}{|\det DT|}$. Thus by change of variables,

$$\begin{aligned} \iint_{\Omega} (y - x) \cos xy \, dA &= \int_1^4 \int_{-4}^{-1} \sqrt{u^2 - 4v} \cos v \cdot \frac{1}{\sqrt{u^2 - 4v}} \, dv \, du \\ &= \int_1^4 \int_{-4}^{-1} \cos v \, dv \, du = \int_1^4 [\sin v]_{-4}^{-1} \, du = 3(\sin(-1) - \sin(-4)) \end{aligned}$$

Example 2.7. Compute $\iint_D e^{-(x^2+y^2)} \, dA$ where $D = \{x^2 + y^2 \leq R^2\}$.

To solve, first note that $f(x, y) = e^{-(x^2+y^2)}$ is C^0 on Ω , thus it is Riemann integrable over Ω . Now we perform a change of variables with $x = r \cos \theta$ and $y = r \sin \theta$. Then $T : \Omega' \rightarrow \Omega$ is defined as $T(r, \theta) = (r \cos \theta, r \sin \theta)$. Then T is a C^1 bijection, where $\Omega' = \{(r, \theta) : 0 \leq r \leq R, 0 \leq \theta \leq 2\pi\}$. Also, $DT = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}$ and $\det DT = r$. Thus by change of variables,

$$\iint_D e^{-(x^2+y^2)} \, dA = \int_0^R \int_0^{2\pi} e^{-r^2} \cdot r \, d\theta \, dr = 2\pi \int_0^R r e^{-r^2} \, dr = \pi(1 - e^{-R^2})$$

Example 2.8 (spherical coordinates). Let $x = p \sin \varphi \cos \theta$, $y = p \sin \varphi \sin \theta$, $z = p \cos \varphi$. Let $\Omega = \{(x, y, z) : x^2 + y^2 + z^2 \leq R^2\}$ and $\Omega' = \{(p, \varphi, \theta) : 0 \leq p \leq a, 0 \leq \varphi \leq \pi, 0 \leq \theta \leq 2\pi\}$. Then for any $f : \Omega \rightarrow \mathbb{R}$ Riemann integrable over Ω ,

$$\iiint_{\Omega} f \, dV = \iiint_{\Omega'} f(p \sin \varphi \cos \theta, p \sin \varphi \sin \theta, p \cos \varphi) \cdot p^2 \sin \varphi \, dp \, d\theta \, d\varphi$$

2.7. IMPROPER INTEGRALS

Definition 27. Suppose $A \subset \mathbb{R}^n$ a region, $f : A \rightarrow \mathbb{R}$, $f(x) \geq 0 \quad \forall x \in A$. Suppose $\forall r > 0$, f is bounded and integrable over $A \cap B_r(0)$. Then we define

$$\int_A f \, dV := \lim_{r \rightarrow \infty} \int_{B_r(0) \cap A} f \, dV$$

If it exists, is is the **improper integral** of f over A .

Definition 28. Consider $\{A_k\}_{k=1}^{\infty}$, $A_k \subset A$, $A_k \uparrow$. Then $\{A_k\}_{k=1}^{\infty}$ are **exhausting regions** of A if

- (1) A_k compact for all k ;
- (2) $\text{Content}(\partial A_k) = 0$;
- (3) $\forall r > 0$, $\exists k_0(r)$ s.t. $B_r(0) \cap A \subset A_k$, $\forall k \geq k_0(r)$.

For example, $\{B_k(0) \cap A\}_{k=1}^{\infty}$ are exhausting regions of A .

Theorem 26. Assume $f \geq 0$ bounded and integrable on $B_r(0) \cap A$, $\forall r > 0$. Then for any exhausting regions $\{A_k\}$ of A ,

$$\lim_{k \rightarrow \infty} \int_{A_k} f \, dV = \lim_{r \rightarrow \infty} \int_{B_r(0) \cap A} f \, dV$$

Example 2.9. Evaluate $\int_0^{\infty} e^{-x^2} \, dx$.

To solve, first note that $f(x) = e^{-x^2}$ is bounded and integrable over $[0, r]$ for all $r > 0$. Thus $\int_0^{\infty} e^{-x^2} \, dx$ exists. Also, $\{[0, k]\}_{k=1}^{\infty}$ are exhausting regions of $[0, \infty)$. Thus,

$$\begin{aligned} \left(\int_0^{\infty} e^{-x^2} \, dx \right)^2 &= \int_0^{\infty} e^{-x^2} \, dx \cdot \int_0^{\infty} e^{-y^2} \, dy = \lim_{k \rightarrow \infty} \iint_{[0, k]^2} e^{-(x^2+y^2)} \, dA \\ &= \lim_{k \rightarrow \infty} \int_0^k \int_0^k e^{-(x^2+y^2)} \, dy \, dx = \lim_{k \rightarrow \infty} \frac{\pi}{2} \int_0^k r e^{-r^2} \, dr = \frac{\pi}{4} \end{aligned}$$

Therefore $\int_0^{\infty} e^{-x^2} \, dx = \frac{\sqrt{\pi}}{2}$.

Definition 29. Given $f : A \rightarrow \mathbb{R}$, define $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$. Then $f = f^+ - f^-$ and $|f| = f^+ + f^-$. If both $\int_A f^+ \, dV$ and $\int_A f^- \, dV$ exist, then f integrable over A and

$$\int_A f \, dV = \int_A f^+ \, dV - \int_A f^- \, dV$$

3. VECTOR FIELDS, CURVES, AND SURFACES

3.1. PATH INTEGRALS/ LINE INTEGRALS

Definition 30. A mapping $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is called $C^1([a, b])$ if $\gamma \in C^1(a, b)$ and

$$\lim_{t \rightarrow a^+} \frac{\gamma(t) - \gamma(a)}{t - a} = \gamma'(a^+) \quad \text{and} \quad \lim_{t \rightarrow b^-} \frac{\gamma(t) - \gamma(b)}{t - b} = \gamma'(b^-)$$

$\gamma : [a, b] \rightarrow \mathbb{R}^n$ is a **parameteric curve** in $\mathbb{R}^n$ if $\gamma \in C^1([a, b])$. Then

$$\gamma(t) = (x_1(t), x_2(t), \dots, x_n(t)) \quad \text{and} \quad \gamma'(t) = (x'_1(t), x'_2(t), \dots, x'_n(t)) \text{ is the tangent vector}$$

Definition 31. Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a continuous map. Then γ is piecewise C^1 if it is C^1 on each subinterval of a partition of $[a, b]$. i.e. it has finitely many "corners"/non-differentiable points.

We say γ is **regular** if $\gamma \in C^1([a, b])$ and $\gamma'(t) \neq 0$ for all $t \in [a, b]$. We say γ is **piecewise regular** if it is regular on each subinterval of a partition of $[a, b]$.

Proposition 7. Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be regular. Then there exists a partition $a = t_0 < t_1 < \dots < t_N = b$ s.t. on each $[t_{i-1}, t_i]$, $i = 1, \dots, N$, there is a j s.t. $t \mapsto x_{j_i}(t)$ is C^1 invertible and for $t \in [t_{i-1}, t_i]$, $\gamma(t) = (x_1(x_{j_i}(t)), \dots, x_{j_{i-1}}(x_{j_i}(t)), x_{j_i}(t), x_{j_{i+1}}(x_{j_i}(t)), \dots, x_n(x_{j_i}(t)))$. i.e. γ is a graph over x_{j_i} .

Definition 32. For each piecewise C^1 curve γ , we define the **length** of γ as

$$L(\gamma) := \sum_{i=1}^n \int_{a_{i-1}}^{a_i} \|\gamma'(t)\| dt = \sum_{i=1}^n \int_{a_{i-1}}^{a_i} \sqrt{(x'_1(t))^2 + \dots + (x'_n(t))^2} dt \quad 6$$

Definition 33 (Reparameterization). Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be C^1 . $\tilde{\gamma} : [c, d] \rightarrow \mathbb{R}^n$ is a **reparameterization** of γ if there exists a C^1 bijection $h : [c, d] \rightarrow [a, b]$ s.t. $\tilde{\gamma}(t) = \gamma(h(t))$.

Note then $\tilde{\gamma}'(t) = h'(t) \cdot \gamma'(h(t))$. Thus, if γ is regular and $h'(t) > 0$ for all t , then $\tilde{\gamma}$ is also regular.

Theorem 27. Suppose $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is a C^1 curve, and $\tilde{\gamma} : [c, d] \rightarrow \mathbb{R}^n$ is a C^1 reparameterization. Then $L(\gamma) = L(\tilde{\gamma})$.

Definition 34 (arc-length parameterization). Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a C^1 regular parameterized curve. Then define

$$s(t) = \int_a^t \|\gamma'(u)\| du \in C^1([a, b]) \quad \text{and} \quad s'(t) = \|\gamma'(t)\| > 0 \quad \forall t \in [a, b]$$

Then $s : [a, b] \rightarrow [0, L]$ is a C^1 monotone increasing bijection where $L = L(\gamma) = \int_a^b \|\gamma'(t)\| dt$. Thus, s has a C^1 inverse $t(s) : [0, L] \rightarrow [a, b]$. Then $\tilde{\gamma}(s) = \gamma(t(s))$ is a reparameterization of γ and is called the **arc-length parameterization** of γ . Note that $\|\tilde{\gamma}'(s)\| = 1$ for all s .

Definition 35. Let f be C^0 in a neighbourhood of γ . The **path integral** of f along γ is defined as

$$\int_{\gamma} f ds := \int_a^b f(\gamma(t)) \cdot \|\gamma'(t)\| dt$$

If we instead use arc-length to parameterize the curve, then

$$\int_{\gamma} f ds = \int_0^L f(\gamma(t(s))) ds$$

by the change of variables formula, since $s'(t) = \|\gamma'(t)\|$

⁶Proof follows from MVT

3.2. VECTOR FIELDS AND LINE INTEGRALS

Definition 36 (Vector Field). A **vector field** on $\Omega \subset \mathbb{R}^n$ is a map $\vec{F} : \Omega \rightarrow \mathbb{R}^n$.

Definition 37. The **total work** done by $\vec{F}$ along γ is defined as

$$\int_{\gamma} \vec{F} \cdot d\vec{s} := \lim_{N \rightarrow \infty} \sum_{i=1}^N \vec{F}(\gamma(t_i)) \cdot \Delta s_i \quad \text{and} \quad = \int_a^b \vec{F}(\gamma(t)) \cdot \gamma'(t) dt \quad \text{assuming the integral exists.}$$

Definition 38 (Line Integral). Let $\gamma \subset \mathbb{R}^n$ be a piecewise smooth path and let $\vec{F}$ be a vector field defined on a neighbourhood of γ . The **line integral** of $\vec{F}$ along γ is defined as

$$\int_{\gamma} \vec{F} \cdot d\vec{s} = \int_a^b \vec{F}(\gamma(t)) \cdot \gamma'(t) dt \quad \text{if } \gamma \text{ is regular.}$$

Note if we let $\vec{T}(t) = \frac{\gamma'(t)}{\|\gamma'(t)\|}$ be the unit tangent field, then

$$\int_{\gamma} \vec{F} \cdot d\vec{s} = \int_{\gamma} \vec{F}(\gamma(t)) \cdot \vec{T}(t) ds \quad \text{where } ds = \|\gamma'(t)\| dt$$

Definition 39. Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a piecewise C^1 path and $\tilde{\gamma} : [c, d] \rightarrow \mathbb{R}^n$ a reparameterization of γ with a bijective C^1 map $h : [c, d] \rightarrow [a, b]$ s.t. $\tilde{\gamma}(t) = \gamma(h(t))$. Then

- (1) $\tilde{\gamma}$ is **orientation-preserving** if $\tilde{\gamma}(c) = \gamma(a)$ and $\tilde{\gamma}(d) = \gamma(b)$;
- (2) $\tilde{\gamma}$ is **orientation-reversing** if $\tilde{\gamma}(c) = \gamma(b)$ and $\tilde{\gamma}(d) = \gamma(a)$.

Lemma 28. Let $\gamma : [a, b] \rightarrow \mathbb{R}^n$ be a piecewise smooth path and $\tilde{\gamma} : [c, d] \rightarrow \mathbb{R}^n$ a reparameterization of γ . Let $\vec{F}$ be a C^0 vector field. Then

$$\int_{\gamma} \vec{F} \cdot d\vec{s} = \epsilon \int_{\tilde{\gamma}} \vec{F} \cdot d\vec{s} \quad \text{where } \epsilon = 1 \text{ if } \tilde{\gamma} \text{ is orientation-preserving and } \epsilon = -1 \text{ otherwise.}$$

3.3. CONSERVATIVE VECTOR FIELDS

Definition 40 (Conservative Vector Field). $\vec{F} \in C^1(\Omega, \mathbb{R}^n)$ is **conservative** if $\exists f \in C^1(\Omega)$ s.t. $\vec{F} = \nabla f$.

Example 3.1. $\vec{F}(x, y) = (x, y)$, $(x, y) \in \mathbb{R}^2$ is the Radial Vector Field. Here, set $f(x, y) = \frac{1}{2}(x^2 + y^2)$. Then $\nabla f = (x, y) = \vec{F}$, so $\vec{F}$ is conservative.

Theorem 29. Suppose $\vec{F}$ is conservative in a neighbourhood of a piecewise smooth path γ . Then

$$\int_{\gamma} \vec{F} \cdot d\vec{s} = f(\gamma(b)) - f(\gamma(a))$$

Corollary 6. If $\vec{F}$ is conservative in a neighbourhood of a closed piecewise smooth path γ , then $\int_{\gamma} \vec{F} \cdot d\vec{s} = 0$.

Theorem 30. Assume $\vec{F} = (F_1, F_2, \dots, F_n) \in C^1(\Omega, \mathbb{R}^n)$ is conservative. Then

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x) \quad \forall i, j = 1, \dots, n, \forall x \in \Omega$$

3.4. POINCARÉ'S LEMMA

Lemma 31. Suppose $g(x, y)$ is continuous defined in $\Omega \times A$, where $\Omega \subset \mathbb{R}^n$ open and $A \subset \mathbb{R}^m$ compact. Assume in addition that $g \in C^1(\Omega \times A)$. Then

$$\frac{\partial}{\partial x_i} \left(\int_A g(x, y) dy \right) = \int_A \frac{\partial}{\partial x_i} g(x, y) dy \quad \forall i = 1, \dots, n$$

Definition 41. $\Omega \subset \mathbb{R}^n$ is **star-shaped** if $\exists p_0 \in \Omega$ s.t. the line $\ell(p_0, x)$ is contained in Ω for all $x \in \Omega$. i.e. $\forall x \in \Omega, \{(1-t)p_0 + tx : t \in [0, 1]\} \subset \Omega$.

Theorem 32 (Poincaré's Lemma). Suppose $\Omega \subset \mathbb{R}^n$ is star shaped and $\vec{F} \in C^1(\Omega, \mathbb{R}^n)$. Then $\vec{F}$ is conservative iff

$$\frac{\partial F_i}{\partial x_j}(x) = \frac{\partial F_j}{\partial x_i}(x) \quad \forall i, j = 1, \dots, n, \forall x \in \Omega$$

3.5. SURFACES AND SURFACE INTEGRALS

Definition 42 (Parameterized Surface). A **parameterized surface** $S \subset \mathbb{R}^3$ is the image of a map $\Phi : D \rightarrow \mathbb{R}^3$,

$$S = \{(x(u, v), y(u, v), z(u, v)) = \Phi(u, v) : (u, v) \in D\}$$

Where $D \subset \mathbb{R}^2$ is some bounded domain. If Φ is C^k , we call S a C^k surface. Note that:

$$\Phi_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right) \quad \text{and} \quad \Phi_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right)$$

are the tangent vectors of S at $\Phi(u, v)$. The tangent plane of S at $\Phi(u, v)$ is the plane spanned by Φ_u and Φ_v .

Definition 43 (Regular Surface). We say S is a **regular surface** if $\vec{\Phi}_u \times \vec{\Phi}_v \neq \vec{0}$ for all $(u, v) \in D$.

Definition 44 (Area of a Surface). Let S be a regular C^1 surface parameterized by $\Phi : D \rightarrow S$. Then the area of S is defined as

$$\text{area}(S) := \iint_D \|\vec{\Phi}_u \times \vec{\Phi}_v\| dA$$

Here,

$$\|\vec{\Phi}_u \times \vec{\Phi}_v\| = \sqrt{\left| \frac{\partial(x, y)}{\partial(u, v)} \right|^2 + \left| \frac{\partial(y, z)}{\partial(u, v)} \right|^2 + \left| \frac{\partial(x, z)}{\partial(u, v)} \right|^2}$$

Definition 45. Let $S = \Phi(D)$ be a C^1 parameterized surface and suppose $f \in C^0(S)$, i.e. f is C^0 in a neighbourhood of S in $\mathbb{R}^3$, i.e. $f \circ \Phi \in C^0(D, \mathbb{R})$. Then the **surface integral** of f over S is defined as

$$\iint_S f dS := \iint_D f(\Phi(u, v)) \cdot \|\vec{\Phi}_u \times \vec{\Phi}_v\| du dv$$

Example 3.2. There is an important special case. Suppose $S = \{(x, y, g(x, y)) : (x, y) \in D \text{ and } g \in C^1(D)\}$. Here $(x, y) = (u, v)$, $\Phi(u, v) = (u, v, g(u, v))$, $\Phi \in C^1(D, \mathbb{R}^3)$. In this case

$$\vec{\Phi}_x = (1, 0, g_x) \quad \text{and} \quad \vec{\Phi}_y = (0, 1, g_y) \quad \text{and} \quad \|\vec{\Phi}_x \times \vec{\Phi}_y\| = \sqrt{1 + g_x^2 + g_y^2}$$

Then $dS = \sqrt{1 + g_x^2 + g_y^2} dx dy$ and so

$$\iint_S f dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} dx dy$$

Example 3.3 (Helicoid). $D = \{0 \leq \theta \leq 2\pi, 0 \leq r \leq 1\}$. $\Phi(r, \theta) = (r \cos \theta, r \sin \theta, \theta)$ is a C^1 parameterization of the helicoid. Then

$$\vec{\Phi}_r = (\cos \theta, \sin \theta, 0) \quad \text{and} \quad \vec{\Phi}_\theta = (-r \sin \theta, r \cos \theta, 1) \quad \text{and} \quad \|\vec{\Phi}_r \times \vec{\Phi}_\theta\| = \sqrt{1 + r^2}$$

Definition 46. Let $S = \Phi(D)$ be a C^1 parameterized surface. We say that S is **regular** if $(\vec{\Phi}_u \times \vec{\Phi}_v)(u, v) \neq \vec{0}$ for all $(u, v) \in D$. If S is regular, then we can define the **unit normal**

$$\vec{n} = \frac{\pm(\vec{\Phi}_u \times \vec{\Phi}_v)}{\|\vec{\Phi}_u \times \vec{\Phi}_v\|}$$

The choice of $\pm$ is called the **orientation** of S . Note that if S is regular, then $\vec{n}$ is well-defined and continuous on S .

Example 3.4. S is the graph of g , i.e. $S = \{(x, y, g(x, y)) : (x, y) \in D \text{ and } g \in C^1(D)\}$. Then

$$\vec{n} = \frac{\pm(-g_x, -g_y, 1)}{\sqrt{1 + g_x^2 + g_y^2}}$$

Definition 47 (Surface Integral of a Vector Field). Let F be a C^0 vector field on an oriented regular parameterized surface. Then

$$\iint_S \vec{F} \cdot d\vec{S} := \iint_D \vec{F}(\Phi(u, v)) \cdot \vec{n} \, ds = \iint_D \vec{F}(\Phi(u, v)) \cdot \frac{\vec{\Phi}_u \times \vec{\Phi}_v}{\|\vec{\Phi}_u \times \vec{\Phi}_v\|} \cdot \|\vec{\Phi}_u \times \vec{\Phi}_v\| \, du \, dv = \iint_D \vec{F}(\Phi(u, v)) \cdot (\vec{\Phi}_u \times \vec{\Phi}_v) \, du \, dv$$

Remark. The definition is independent of the choice of parameterization, modulo the choice of orientation.

Definition 48 (reparameterization of a surface). Let $\Phi : D \subset \mathbb{R}^2 \rightarrow S$ be a regular C^1 parameterization, $\tilde{\Phi} : \tilde{D} \subset \mathbb{R}^2 \rightarrow S$ another regular C^1 parameterization. We say $\tilde{\Phi}$ is a **reparameterization** of Φ if there exists a C^1 bijection $h : \tilde{D} \rightarrow D$ s.t. $\tilde{\Phi}(u, v) = \Phi(h(u, v))$. Then $\tilde{\Phi}$ is **orientation-preserving** (+) if

$$\frac{\vec{\Phi}_u \times \vec{\Phi}_v(h(s, t))}{\|\vec{\Phi}_u \times \vec{\Phi}_v\| \cdot (h(s, t))} = \frac{\tilde{\vec{\Phi}}_s \times \tilde{\vec{\Phi}}_t(s, t)}{\|\tilde{\vec{\Phi}}_s \times \tilde{\vec{\Phi}}_t\|(s, t)} \quad \forall (s, t) \in \tilde{D}$$

and **orientation-reversing** (-) if

$$\frac{\vec{\Phi}_u \times \vec{\Phi}_v(h(s, t))}{\|\vec{\Phi}_u \times \vec{\Phi}_v\| \cdot (h(s, t))} = -\frac{\tilde{\vec{\Phi}}_s \times \tilde{\vec{\Phi}}_t(s, t)}{\|\tilde{\vec{\Phi}}_s \times \tilde{\vec{\Phi}}_t\|(s, t)} \quad \forall (s, t) \in \tilde{D}$$

Lemma 33. Let $\tilde{\Phi}$ and Φ be as above with $h : D \rightarrow \tilde{D}$ such that $h(s, t) = (u(s, t), v(s, t)) \in C^1(\tilde{D}, D)$. Then h is orientation-preserving iff $\det\left(\frac{\partial(u, v)}{\partial(s, t)}\right) > 0$, and h is orientation-reversing iff $\det\left(\frac{\partial(u, v)}{\partial(s, t)}\right) < 0$.

Proposition 8. Let S be an oriented C^1 regular parameterized surface $S = \Phi(D)$ and $\vec{F} \circ \Phi \in C^0(D, \mathbb{R}^3)$. Then $\iint_S \vec{F} \cdot d\vec{S}$ is well defined independent of the choice of parameterization, modulo the choice of orientation, (+) or (-).

Corollary 7. $S = \Phi(D)$ an oriented regular parameterized surface. Then

$$\iint_{S^\mp} \vec{F} \cdot d\vec{S} = - \iint_{S^\pm} \vec{F} \cdot d\vec{S}$$

3.6. GREEN'S THEOREM

Definition 49 (Regular Surface). $S \subset \mathbb{R}^3$ is a **regular surface** with C^1 boundary if $S \subset \mathbb{R}^3$ is compact and if $\forall p \in S, \exists r, \delta > 0, B_r(p) \subset \mathbb{R}^3$, and $D = \{-\delta < u < \delta, -\delta < v < \delta\} \subset \mathbb{R}^2$ and a regular map $\Phi : D \rightarrow \mathbb{R}^3$ such that either

- (1) $S \cap B_r(p) = \Phi(D)$;
- (2) $S \cap B_r(p) = \Phi(D^+)$, where $D^+ = \{-\delta < u < \delta, 0 \leq v < \delta\}$

Points in (1) are called **interior points** of S and points in (2) are called **boundary points** of S . $\partial S := \{\text{all boundary points of } S\}$

Example 3.5 (orientability). Let $S_+^2 = \{x^2 + y^2 + z^2 = 1, z \geq 0\}$. We say that a regular surface S is **orientable** if there exists a unit normal $\vec{n}(p)$ at each $p \in S$ with $\vec{n} \in C^0(S, \mathbb{R}^3)$.

Definition 50. We say a bounded planar domain $D \subset \mathbb{R}^2$ is (+) **positively oriented** if ∂D is oriented counter-clockwise, i.e. if the unit normal is $\vec{k} = (0, 0, 1)$.

Theorem 34 (Green's Theorem). Let $D \subseteq \mathbb{R}^2$ be a domain with piecewise C^1 boundary ∂D , and $\vec{F} = (P, Q) : D \rightarrow \mathbb{R}^2$ a C^1 vector field on D . Then

$$\int_{\partial D} \vec{F} \cdot d\vec{s} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Corollary 8.

$$\int_{\partial D} x \, dy = - \int_{\partial D} y \, dx = \frac{1}{2} \int_{\partial D} x \, dy - y \, dx = \text{area}(D)$$

Example 3.6. Compute the area of the ellipse $\left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}$ using Green.

To solve, notice the boundary ∂D is the ellipse $\left\{ \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right\}$. We can parameterize ∂D by $\gamma(t) = (a \cos t, b \sin t)$, $t \in [0, 2\pi]$. Then

$$\int_{\partial D} x \, dy = \int_0^{2\pi} a \cos t \cdot b \cos t \, dt = ab \int_0^{2\pi} \cos^2 t \, dt = ab\pi$$

Corollary 9. Let $D = D_1 \setminus D_2$ and $\partial D_1, \partial D_2$ piecewise C^1 with orientations induced from $D_1 + D_2$. Then if $\vec{F} = (P, Q)$ is a C^1 vector field on D with $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 0$ on D , then

$$\int_{\partial D_1} \vec{F} \cdot d\vec{s} = \int_{\partial D_2} \vec{F} \cdot d\vec{s}$$

Theorem 35. If $\vec{F} = (P, Q) \in C^1$, then

$$\int_{\partial D} \vec{F} \cdot \vec{n} \, ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA$$

3.7. HARMONIC FUNCTIONS

Definition 51 (Harmonic Function). Let $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ be the Laplacian. $u \in C^2(D)$ is **harmonic** if $\Delta u = 0$ on D .

Corollary 10. u harmonic on domain $D \subset \mathbb{R}^2$ with piecewise C^1 boundary ∂D . Then

$$\iint_D \Delta u \, dA = \int_{\partial D} \Delta u \cdot \vec{n} \, ds = 0$$

Corollary 11 (Green's Identities). Let $\varphi, \psi \in C^2(D)$, ∂D piecewise C^1 with positive orientation. Then

•

$$\iint_D \varphi \Delta \psi \, dA = - \int_{\partial D} \nabla \varphi \cdot \nabla \psi \, dA + \iint_{\partial D} \varphi \nabla \psi \cdot \vec{n} \, dA$$

•

$$\iint_D [\varphi \Delta \psi - \psi \Delta \varphi] \, dA = \int_{\partial D} (\varphi \nabla \psi - \psi \nabla \varphi \cdot \vec{n}) \, ds$$

3.8. STOKES' THEOREM

Definition 52. Let $\vec{F} = (F_1, F_2, F_3) : \Omega \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a C^1 vector field. Then the **curl** of $\vec{F}$ is defined as

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \det \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} \\ F_1 & F_2 & F_3 \end{bmatrix} = \left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}, \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}, \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2} \right)$$

Theorem 36 (Stokes). Let S be a compact regular oriented surface with $(+)$ orientation and piecewise C^1 boundary ∂S . Let $F : S \rightarrow \mathbb{R}^3$ be a C^1 vector field. Then

$$\iint_S \text{curl } \vec{F} \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

Remark. Note,

$$\iint_S \text{curl } \vec{F} \cdot d\vec{S} = \iint_S \text{curl } \vec{F} \cdot \vec{n} \, dS$$

Example 3.7. $\vec{F} = (P, Q, 0)$. Then $F_3 = 0$ and $P = P(x, y), Q = Q(x, y)$, so $\text{curl } \vec{F} = \left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)$. Then we can see Green's theorem is a really just a special case of Stokes' theorem, since

$$\left(0, 0, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) \cdot \vec{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$$

Corollary 12. Suppose S_1, S_2 are two oriented surfaces with $\partial S_1 = \partial S_2 = \gamma$, and suppose S_1 and S_2 induce opposite orientations on γ . Then, for any $\vec{F} \in C^1$,

$$\iint_{S_1} \text{curl } \vec{F} \cdot d\vec{S} = - \iint_{S_2} \text{curl } \vec{F} \cdot d\vec{S}$$

Corollary 13. If S is a compact, oriented surface with $\partial S = \emptyset$ and $\vec{F} \in C^1(S)$, then

$$\iint_S \text{curl } \vec{F} \cdot d\vec{S} = 0$$

Corollary 14. Suppose $\vec{F} = \nabla F$ for some $f \in C^2$ and S is an oriented surface with $\partial S = \gamma_1 - \gamma_2$. Then,

$$\int_{\gamma_1} \vec{F} \cdot d\vec{s} = \int_{\gamma_2} \vec{F} \cdot d\vec{s}$$

Example 3.8. Let $C = \{y + z = 2\} \cap \{x^2 + y^2 = 1\}$ with $(+)$ orientation. Let $\vec{F} = (-y^2, x, z^2)$. Evaluate $\int_C \vec{F} \cdot d\vec{s}$.

To solve, by Stokes

$$\int_C \vec{F} \cdot d\vec{s} = \iint_S \text{curl } \vec{F} \cdot d\vec{S}$$

Here $\text{curl } \vec{F} = (0, 0, 1 + 2y)$ and $\vec{n} = \frac{(0, 1, 1)}{\sqrt{2}}$. Therefore,

$$\int_C \vec{F} \cdot d\vec{s} = \iint_D (0, 0, 1 + 2y) \cdot (0, 1, 1) dA = \iint_D (1 + 2y) dA$$

To solve this integral, we make the change of variables $x = r \cos \theta, y = r \sin \theta, D = \{0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$. Then we get

$$= \int_0^{2\pi} \int_0^1 (1 + 2r \sin \theta) r dr d\theta = \int_0^{2\pi} \left(\frac{1}{2} + \frac{2}{3} \sin \theta \right) d\theta = \pi$$

3.9. DIVERGENCE THEOREM

Theorem 37 (Divergence Theorem). Let $W \subset \mathbb{R}^3$ be a bounded solid region in $\mathbb{R}^3$ with piecewise smooth boundary $S = \partial W$ with positive orientation and $\vec{F} \in C^1(W, \mathbb{R}^3)$. Then

$$\iiint_W \text{div } \vec{F} dV = \iint_S \vec{F} \cdot d\vec{S}$$

Here, $\text{div } \vec{F} := \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$

Corollary 15. Under the same assumptions as above, for any $f, g \in C^2(W)$,

$$\iiint_W [g\Delta f - f\Delta g] dV = \iint_{\partial W} [g\nabla f \cdot d\vec{S} - f\nabla g \cdot d\vec{S}]$$

Corollary 16. Suppose $W, W_1, W_2, \dots, W_k \subset \mathbb{R}^n$ are regions as in the divergence theorem with

- $W_i \cap W_j = \emptyset$ for all $i \neq j$;
- $W_j \subset W^{\text{int}}, p_j \in W_j^{\text{int}}$ for all $j = 1, \dots, k$;

for some points $p_1, \dots, p_k$. If $\vec{F} \in C^1(W \setminus \{p_1, \dots, p_k\})$ and $\text{div } \vec{F} = 0$ for all $x \in W \setminus \{p_1, \dots, p_k\}$, then

$$\iint_{\partial W} \vec{F} \cdot d\vec{S} = \sum_{j=1}^k \iint_{\partial W_j} \vec{F} \cdot d\vec{S}$$

Example 3.9 (Gauss law). Suppose $W \subset \mathbb{R}^3$ as in the divergence theorem and $\vec{F} = (x, y, z)$ and $r = \sqrt{x^2 + y^2 + z^2}$. Then, if $(0, 0, 0) \notin \partial W$,

$$\iint_{\partial W} \frac{\vec{F}}{r^3} \cdot d\vec{S} = \begin{cases} 4\pi & \text{if } (0, 0, 0) \in W^{\text{int}} \\ 0 & \text{otherwise} \end{cases}$$

To show this, set $\vec{F} = \frac{\vec{r}}{r^3}$. Then $\text{div } \vec{F} = 0, \vec{r} \in \mathbb{R}^3 \setminus (0, 0, 0)$. SO if $(0, 0, 0) \notin W, \vec{F} \in C^1(W)$ and by the divergence theorem, $\iint_{\partial W} \frac{\vec{r}}{r^3} \cdot d\vec{S} = 0$. If $(0, 0, 0) \in W$, pick $\delta > 0$ small so that $B_\delta(0) \subset W^{\text{int}}$. Let $\tilde{W} = W \setminus B_\delta(0)$. By the most recent corollary,

$$\iint_{\partial W} \frac{\vec{r}}{r^3} \cdot d\vec{S} = \iint_{\partial B_\delta(0)} \frac{\vec{r}}{r^3} \cdot d\vec{S} = \iint_{\partial B_\delta(0)} \frac{\vec{r}}{\delta^3} \cdot d\vec{S} = \frac{1}{\delta^3} \iint_{\partial B_\delta(0)} \vec{r} \cdot d\vec{S} = 4\pi$$

3.10. DIVERGENCE THEOREM APPLICATIONS

Example 3.10. Compute $\iint_S \vec{F} \cdot d\vec{S}$ where $\vec{F} = (3x + z^{77}, y^2 - \sin x^2 z, xz + ye^{x^3})$, $S = \partial W$ where $W = \{0 \leq x \leq 1, 0 \leq y \leq 3, 0 \leq z \leq 2\}$.

To solve, we can use the divergence theorem. We have $\operatorname{div} F = 3 + 2y + x$. Then by the divergence theorem,

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iiint_W (3 + 2y + x) \, dV = \int_0^1 \int_0^3 \int_0^2 (3 + 2y + x) \, dz \, dy \, dx \\ &= \int_0^1 \int_0^3 2(3 + 2y + x) \, dy \, dx \\ &= \int_0^1 2(9 + 9 + 3x) \, dx \\ &= \int_0^1 36 + 6x \, dx = 39 \end{aligned}$$